

CA – 271 –VSC Lab Course Python Programming

Slip 1

1) Write a Python Program to Calculate the Area of a Rectangle and Circle.

```
import math

length = float(input("Enter the length of the rectangle: "))
width = float(input("Enter the width of the rectangle: "))
rectangle_area = length * width

print("Area of Rectangle =", rectangle_area)

radius = float(input("Enter the radius of the circle: "))
circle_area = math.pi * radius * radius

print("Area of Circle =", circle_area)
```

2) Write a Python program to Remove duplicate characters and Vowels from a string and display the modified string.

```
string = input("Enter a string: ")
vowels = "aeiouAEIOU"
result = ""
seen = set()

for char in string:
    if char not in vowels and char not in seen:
        result += char
```

```
seen.add(char)
```

```
print("Modified string:", result)
```

OR

2) Write a Python program to get the maximum and minimum value in a dictionary.

```
my_dict = {  
    "a": 45,  
    "b": 12,  
    "c": 78,  
    "d": 34,  
    "e": 9  
}  
  
max_value = max(my_dict.values())  
min_value = min(my_dict.values())  
print("Dictionary:", my_dict)  
print("Maximum value:", max_value)  
print("Minimum value:", min_value)
```

Slip 2

1) Write a python program to replace special symbol in string with # character.

```
import re
```

```
string = input("Enter a string: ")
```

```
# Replace special symbols with '#'
modified_string = re.sub(r'^a-zA-Z0-9 ', '#', string)

print("Modified string:", modified_string)
```

2) Write a Python program to find the repeated items of a tuple and remove repeated items of a tuple and display updated tuple.

```
t = (1, 2, 3, 2, 4, 5, 1, 6, 3)

print("Original Tuple:", t)

repeated = set()
seen = set()

for item in t:
    if item in seen:
        repeated.add(item)
    else:
        seen.add(item)

print("Repeated items:", tuple(repeated))

updated_tuple = tuple(set(t))

print("Updated Tuple:", updated_tuple)
```

OR

2) Write a Python Program to Print the pattern as below

```
*  
* *  
* * *
```

```
n = 3  
for i in range(1, n + 1):  
    for j in range(i):  
        print("*", end=" ")  
    print()
```

Slip 3

1) Write a Python program to find the length of a tuple.

```
t = (10, 20, 30, 40, 50)  
length = len(t)  
print("Tuple:", t)  
print("Length of tuple:", length)
```

2) Write a Python program to Display multiplication table of a number.

```
num = int(input("Enter a number: "))  
  
for i in range(1, 11):  
    print(num, "x", i, "=", num * i)
```

OR

2) Write a Python program to: accept a sentence and Display total number of words and each word with its starting index.

```
sentence = input("Enter a sentence: ")

words = sentence.split()

print("Total number of words:", len(words))

index = 0

for word in words:
    print(word, "starts at index", sentence.index(word, index))
    index = sentence.index(word, index) + len(word)
```

Slip 4

1) Write a Python Program to Calculate the Sum and Average of Numbers in a Given List.

```
numbers = [10, 20, 30, 40, 50]
```

```
total = sum(numbers)
```

```
average = total / len(numbers)
```

```
print("Numbers:", numbers)
```

```
print("Sum:", total)
```

```
print("Average:", average)
```

2) Write a Python program to create a dictionary of keys x, y, and z where each key has as value a list from 11-20, 21-30, and 31-40 respectively. Access the fifth value of each key from the dictionary.

```
my_dict = {
    "x": list(range(11, 21)),
    "y": list(range(21, 31)),
```

```
"z": list(range(31, 41))
}

print("Dictionary:", my_dict)
print("Fifth value of x:", my_dict["x"][4])
print("Fifth value of y:", my_dict["y"][4])
print("Fifth value of z:", my_dict["z"][4])
```

OR

2) Write a Python program to accept two strings and Check whether both strings are equal or not. Display appropriate message (without using ==)

```
str1 = input("Enter first string: ")
str2 = input("Enter second string: ")

if len(str1) != len(str2):
    print("Strings are not equal")
else:
    equal = True
    for i in range(len(str1)):
        if str1[i] != str2[i]:
            equal = False
            break
    if equal:
        print("Strings are equal")
    else:
        print("Strings are not equal")
```

Slip 5

1) Write a Python program to convert a list to a tuple.

```
my_list = [1, 2, 3, 4, 5]

my_tuple = tuple(my_list)

print("List:", my_list)
print("Tuple:", my_tuple)
```

2) Write a Python program to Accept Three Numbers and find Minimum and Maximum of these Numbers.

```
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))
num3 = float(input("Enter third number: "))

minimum = min(num1, num2, num3)
maximum = max(num1, num2, num3)

print("Minimum number:", minimum)
print("Maximum number:", maximum)
```

OR

2) Write a Python function that accepts a string and calculate the number of upper case letters and lower case letters.

```
def count_case(s):
    upper = sum(1 for c in s if c.isupper())
    lower = sum(1 for c in s if c.islower())
    return upper, lower
```

```
text = input("Enter a string: ")
upper_count, lower_count = count_case(text)

print("Uppercase letters:", upper_count)
print("Lowercase letters:", lower_count)
```

Slip 6

1) Write a Python program to remove duplicates from a list.

```
my_list = [1, 2, 3, 2, 4, 1, 5, 3]

unique_list = list(set(my_list))

print("Original List:", my_list)
print("List after removing duplicates:", unique_list)
```

2) Write a Python function that accepts a string and calculate the number of upper case letters and lower case letters.

```
def count_letters(s):
    upper_count = 0
    lower_count = 0
    for char in s:
        if char.isupper():
            upper_count += 1
        elif char.islower():
            lower_count += 1
    return upper_count, lower_count
```



```
text = input("Enter a string: ")
upper, lower = count_letters(text)

print("Uppercase letters:", upper)
print("Lowercase letters:", lower)
```

OR

2) Write a Python program to sort (ascending and descending) a dictionary by value.

```
my_dict = {"a": 45, "b": 12, "c": 78, "d": 34}

# Ascending order
asc_sorted = dict(sorted(my_dict.items(), key=lambda item: item[1]))

# Descending order
desc_sorted = dict(sorted(my_dict.items(), key=lambda item: item[1], reverse=True))

print("Original Dictionary:", my_dict)
print("Ascending order by value:", asc_sorted)
print("Descending order by value:", desc_sorted)
```

Slip 7

1)Write a Python program to find maximum and the minimum value in a set.

```
my_set = {15, 42, 7, 23, 89, 5}

max_value = max(my_set)
min_value = min(my_set)

print("Set:", my_set)
```

```
print("Maximum value:", max_value)
print("Minimum value:", min_value)
```

2) Write a Python program to mask a string such that:

- First and last characters remain same
- All middle characters are replaced by * Example Enter a string: programming Masked string: p*****g

```
string = input("Enter a string: ")

if len(string) <= 2:
    masked = string
else:
    masked = string[0] + '*' * (len(string) - 2) + string[-1]

print("Masked string:", masked)
```

OR

2) Write a Python program to accept two numbers and perform Bitwise AND, OR, NOT and XOR operations on Them.

```
num1 = int(input("Enter first number: "))
num2 = int(input("Enter second number: "))

bitwise_and = num1 & num2
bitwise_or = num1 | num2
bitwise_xor = num1 ^ num2
bitwise_not_num1 = ~num1
bitwise_not_num2 = ~num2

print(f"Bitwise AND: {bitwise_and}")
```

```
print(f"Bitwise OR: {bitwise_or}")
print(f"Bitwise XOR: {bitwise_xor}")
print(f"Bitwise NOT of {num1}: {bitwise_not_num1}")
print(f"Bitwise NOT of {num2}: {bitwise_not_num2}")
```

Slip 8

1) Write a Python program to merge two Python dictionaries.

```
dict1 = {"a": 1, "b": 2, "c": 3}
dict2 = {"d": 4, "e": 5, "f": 6}

merged_dict = {**dict1, **dict2}

print("Dictionary 1:", dict1)
print("Dictionary 2:", dict2)
print("Merged Dictionary:", merged_dict)
```

2)2 Write a Python program to check whether a string Starts with a capital letter, Ends with a digit and contain at least one special character. Display appropriate messages.

```
import string

s = input("Enter a string: ")

special_characters = string.punctuation

if s[0].isupper():
    print("String starts with a capital letter.")
else:
    print("String does not start with a capital letter.")
```

```
if s[-1].isdigit():
    print("String ends with a digit.")
else:
    print("String does not end with a digit.")

if any(char in special_characters for char in s):
    print("String contains at least one special character.")
else:
    print("String does not contain any special character.")
```

OR

2) Write a Python program to accepts marks of 3 subjects (Out of 100) and calculate the Total and Average of the marks. Also find the grade of a students.

```
marks1 = float(input("Enter marks for subject 1: "))
marks2 = float(input("Enter marks for subject 2: "))
marks3 = float(input("Enter marks for subject 3: "))
```

```
total = marks1 + marks2 + marks3
average = total / 3
```

```
print("Total marks:", total)
print("Average marks:", average)
```

```
if average >= 90:
    grade = "A"
elif average >= 80:
    grade = "B"
elif average >= 70:
    grade = "C"
elif average >= 60:
```

```
    grade = "D"
else:
    grade = "F"

print("Grade:", grade)
```

Slip 9

1) Write a Python program to get the largest and smallest number from a list.

```
numbers = [10, 45, 23, 67, 12, 89, 5]

largest = max(numbers)

smallest = min(numbers)

print("List of numbers:", numbers)
print("Largest number:", largest)
print("Smallest number:", smallest)
```

2) Write a Python function that accepts a string and calculate the number of vowels and consonants of a string and display them.

```
def count_vowels_consonants(text):
    vowels = "aeiouAEIOU"
    v_count = 0
    c_count = 0

    for ch in text:
        if ch.isalpha():
            if ch in vowels:
                v_count += 1
            else:
                c_count += 1

    print("Number of vowels:", v_count)
    print("Number of consonants:", c_count)
```

```
string = input("Enter a string: ")
count_vowels_consonants(string)
```

OR

2) Write a Python Program to display whether a given number is Armstrong or Not.

```
num = int(input("Enter a number: "))
sum = 0
temp = num

while temp > 0:
    digit = temp % 10
    sum = sum + digit ** 3
    temp = temp // 10

if sum == num:
    print(num, "is an Armstrong number")
else:
    print(num, "is not an Armstrong number")
```

Slip 10

1) Write a Python program to define a function that counts the number of digits in a given number.

```
def count_digits(num):
    count = 0

    while num != 0:
        num = num // 10
        count += 1
```

```
    return count

number = int(input("Enter a number: "))

digits = count_digits(number)

print("Number of digits:", digits)
```

2) Write a Python program to find number is Armstrong or not.

```
num = int(input("Enter a number: "))
temp = num
sum = 0

while num > 0:
    digit = num % 10
    sum = sum + digit ** 3
    num = num // 10

if sum == temp:
    print(temp, "is an Armstrong number")
else:
    print(temp, "is not an Armstrong number")
```

OR

2) Write a Python program to count the number of strings where the string length is 3 or more and the first and last character are same from a given list of strings.

```
def count_strings(words):
    count = 0

    for word in words:
        if len(word) >= 3 and word[0] == word[-1]:
            count += 1

    return count
```

```
string_list = ['abc', 'xyz', 'aba', '1221', 'aa', 'level']
```

```
result = count_strings(string_list)
```

```
print("Number of matching strings:", result)
```

Slip 11

1) Write a Python program to remove duplicates from a list and display resultant list.

```
numbers = [10, 20, 30, 20, 40, 10, 50]
```

```
result = list(set(numbers))
```

```
print("Original List:", numbers)
```

```
print("List after removing duplicates:", result)
```

2) Write a Python program to accept a sentence, Count number of words and Find the longest and shortest word.

```
sentence = input("Enter a sentence: ")
```

```
words = sentence.split()
```

```
count = len(words)
```

```
longest = max(words, key=len)
```

```
shortest = min(words, key=len)
```

```
print("Number of words:", count)
```

```
print("Longest word:", longest)
```

```
print("Shortest word:", shortest)
```

OR

2) Write a Python program to print all Odd numbers between 1 and 50 using a while loop.

```
num = 1

while num <= 50:
    if num % 2 != 0:
        print(num)
    num += 1
```

Slip 12

1) Write a python program to arrange string characters such that lowercase letters should come first .

```
string = input("Enter a string: ")

lowercase = ""
uppercase = ""

for ch in string:
    if ch.islower():
        lowercase += ch
    else:
        uppercase += ch

result = lowercase + uppercase

print("Rearranged string:", result)
```

2) Write a Python program to check whether a given number is Divisible by 5 and 7 as well as it is Even or Odd.

```
num = int(input("Enter a number: "))
```

```

if num % 5 == 0 and num % 7 == 0:
    print(num, "is divisible by both 5 and 7")
else:
    print(num, "is not divisible by both 5 and 7")

if num % 2 == 0:
    print(num, "is an Even number")
else:
    print(num, "is an Odd number")

```

OR

2) Write a Python program to accept a list of integers and Count the number of even and odd elements and display all even elements in one list and odd elements in another list.

```

numbers = list(map(int, input("Enter numbers separated by space: ").split()))

even_list = []
odd_list = []

for num in numbers:
    if num % 2 == 0:
        even_list.append(num)
    else:
        odd_list.append(num)

print("Even numbers count:", len(even_list))
print("Odd numbers count:", len(odd_list))
print("Even numbers list:", even_list)
print("Odd numbers list:", odd_list)

```

Slip 13

1) Python Program to Swap Two Numbers.

```

a = 5
b = 10

print("Before swapping:")
print("a =", a)
print("b =", b)

```

```
temp = a
a = b
b = temp

print("After swapping:")
print("a =", a)
print("b =", b)
```

2) Write a python program to find word in string which contain both digit and number.

```
string = input("Enter a string: ")

words = string.split()

print("Words containing both letters and digits:")

for word in words:
    has_letter = False
    has_digit = False

    for ch in word:
        if ch.isalpha():
            has_letter = True
        if ch.isdigit():
            has_digit = True

    if has_letter and has_digit:
        print(word)
```

OR

2) Write a Python Program to Remove All Occurrences of an Element from a Given List.

```
numbers = [10, 20, 30, 20, 40, 20, 50]

element = int(input("Enter element to remove: "))

numbers = [x for x in numbers if x != element]

print("List after removing element:", numbers)
```

Slip 14

1) Write a Python Program to Interchange First and Last Elements of in a List .

```
numbers = [10, 20, 30, 40, 50]
```

```
print("Original List:", numbers)

# Swapping first and last elements
numbers[0], numbers[-1] = numbers[-1], numbers[0]

print("List after swapping first and last elements:", numbers)
```

2) Write a Python program to accept two strings and find common characters between the two strings also check whether both strings are equal or not.

```
str1 = input("Enter first string: ")
str2 = input("Enter second string: ")

common = set(str1) & set(str2)

print("Common characters between the two strings:", common)

if str1 == str2:
    print("Both strings are equal")
else:
    print("Both strings are not equal")
```

OR

2) Write a Python Program to Print the Fibonacci sequence .

```
n = int(input("Enter number of terms: "))

a = 0
b = 1

for i in range(n):
    print(a, end=" ")
    c = a + b
    a = b
    b = c
```

Slip 15

1) Write a Python Program to Square Each Element of the List and Print List in Reverse Order.

```
lst = [int(x) for x in input("Enter elements: ").split()]

sq = []
for i in lst:
    sq.append(i*i)

print("Squared List:", sq)
print("Reverse Order:", sq[::-1])
```

2) Write a Python Program to Find the Sum of Natural Numbers

```
n = int(input("Enter a number: "))

s = 0
i = 1

while i <= n:
    s = s + i
    i = i + 1

print("Sum of natural numbers:", s)
```

OR

2) Write a Python program to accept a sentence and Count number of words also Display the sentence in reverse word order.

```
s = input("Enter a sentence: ")

words = s.split()

print("Number of words:", len(words))
print("Sentence in reverse word order:", " ".join(words[::-1]))
```

Slip 16

1) Write a Python Program to Check whether the year is Leap Year or Not.

```
year = int(input("Enter a year: "))

if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
    print("Leap Year")
```

```
else:  
    print("Not a Leap Year")
```

2) Write a Python program to Remove duplicate characters from a string also replace every space with an underscore _ and Display the modified string.

```
s = input("Enter a string: ")  
  
s = s.replace(" ", "_")  
  
result = ""  
for ch in s:  
    if ch not in result:  
        result += ch  
  
print("Modified string:", result)
```

OR

2) Write a Python Program to Square Each Element of the List and Print List in Reverse Order.

```
lst = [int(x) for x in input("Enter elements: ").split()]  
  
sq = [i*i for i in lst]  
  
print("Squared List:", sq)  
print("Reverse Order:", sq[::-1])
```

Slip 17

1) Write a python program to check whether the string is Palindrome or not.

```
s = input("Enter a string: ")  
  
if s == s[::-1]:  
    print("Palindrome")  
else:  
    print("Not a Palindrome")
```

2) Write a python program that converts a decimal number to binary number using function.

```
def decimal_to_binary(n):  
    b = ""  
    while n > 0:  
        b = str(n % 2) + b  
        n = n // 2  
    return b
```

```
num = int(input("Enter a decimal number: "))
print("Binary number:", decimal_to_binary(num))
```

OR

2) Write a python program to sort the tuple
For example: Tuple=(2, 4,6, 1, 4, 7.8, 2.7)
Output: (1,2,2.7,4,4,6,7.8)

```
t = (2, 4, 6, 1, 4, 7.8, 2.7)

sorted_tuple = tuple(sorted(t))

print(sorted_tuple)
```

Slip 18

1) Write a Python program to find the Maximum of three numbers.

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
c = int(input("Enter third number: "))

if a >= b and a >= c:
    print("Maximum number:", a)
elif b >= a and b >= c:
    print("Maximum number:", b)
else:
    print("Maximum number:", c)
```

2) Write a Python program to check whether an element exists within a tuple.

```
t = (10, 20, 30, 40, 50)

x = int(input("Enter element to search: "))

if x in t:
    print("Element exists in the tuple")
else:
    print("Element does not exist in the tuple")
```

OR

2) Write a Python program to find the length of a set.

```
s = {10, 20, 30, 40, 50}

print("Length of the set:", len(s))
```

Slip 19

1) Write a Python Program to Check accepted number is Prime or not.

```
n = int(input("Enter a number: "))

for i in range(2, n):
    if n % i == 0:
        print("Not a Prime Number")
        break
else:
    if n > 1:
        print("Prime Number")
    else:
        print("Not a Prime Number")
```

2) Write a Python program to combine two dictionary adding values for common keys.

Sample Dictionary: d1={'a':100,'b':200,'c':300} d2={'a':300,'b':200,'d':400}

Sample output: Counter({'a':400, 'b':400,'d':400,'c':300})

```
from collections import Counter

d1 = {'a':100,'b':200,'c':300}
d2 = {'a':300,'b':200,'d':400}

result = Counter(d1) + Counter(d2)

print(result)
```

OR

2) Write a Python program to convert a tuple to a string and display it in reverse manner.

```
t = ('P', 'Y', 'T', 'H', 'O', 'N')
```



```
s = "".join(t)

print("String:", s)
print("Reverse String:", s[::-1])
```

Slip 20

1) Write a python program to Reverse words in a given String.

```
s = input("Enter a string: ")

words = s.split()

print("Reversed words:", " ".join(words[::-1]))
```

2) Write a Python program that removes all even numbers from a specified list and displays the remaining elements.

```
lst = [int(x) for x in input("Enter elements: ").split()]

for i in lst[:]:
    if i % 2 == 0:
        lst.remove(i)

print(lst)
```

OR

3) Write a Python program that will print all prime numbers from a specified range.

```
start = int(input("Enter start number: "))
end = int(input("Enter end number: "))

for n in range(start, end + 1):
    if n > 1:
        for i in range(2, n):
            if n % i == 0:
                break
        else:
            print(n, end=" ")
```

Slip 21

1) Write a Python function to accept Three numbers and find the Minimum of them.

```

def minimum(a, b, c):
    if a <= b and a <= c:
        return a
    elif b <= a and b <= c:
        return b
    else:
        return c

x = int(input("Enter first number: "))
y = int(input("Enter second number: "))
z = int(input("Enter third number: "))

print("Minimum number:", minimum(x, y, z))

```

2) Write a Python program to find symmetric difference of two sets.

```

a = {int(x) for x in input("Enter elements of first set: ").split()}
b = {int(x) for x in input("Enter elements of second set: ").split()}

print("Symmetric Difference:", a ^ b)

```

OR

2) Write a Python program to find the student with the highest marks from a dictionary.

```

marks = {"Amit": 85, "Riya": 92, "John": 78, "Neha": 88}

top_student = max(marks, key=marks.get)

print("Student with highest marks:", top_student)
print("Marks:", marks[top_student])

```

Slip 22

1) Write a Python Program to Find the Factorial of a Number.

```

n = int(input("Enter a number: "))

fact = 1

for i in range(1, n + 1):
    fact = fact * i

print("Factorial:", fact)

```

**2) Write a Python program to generate and print a dictionary that contains a number (Between 1 and n) in the form (x, x*x). Sample Dictionary (n=5)
Expected Output : {1:1,2: 4, 3: 9, 4:16, 5: 25}**

```
n = int(input("Enter a number: "))  
  
d = {x: x*x for x in range(1, n+1)}  
  
print(d)
```

OR

2) Write a python program to convert lowercase letters of a string to uppercase and vice versa.

```
s = input("Enter a string: ")  
print(s.swapcase())
```

Slip 23

1) Write a Python program to get the 4th element from front and 6th element from last of a tuple.

```
t = (10, 20, 30, 40, 50, 60, 70, 80)  
  
fourth_from_front = t[3]  
sixth_from_last = t[-6]  
  
print("4th element from front:", fourth_from_front)  
print("6th element from last:", sixth_from_last)
```

2) Write a Python function to print sum of all the numbers in a list.

```
def sum_of_list(lst):  
    return sum(lst)  
  
numbers = [int(x) for x in input("Enter numbers: ").split()]  
print("Sum of all numbers:", sum_of_list(numbers))
```

OR

2) Write a program to find the factorial of a number using a loop.

```
n = int(input("Enter a number: "))
```

```
factorial = 1
for i in range(1, n + 1):
    factorial *= i

print("Factorial:", factorial)
```

Slip 24

1) Write a Python program to split a list into two lists: One containing positive numbers and another containing negative numbers.

```
lst = [int(x) for x in input("Enter numbers: ").split()]

positive = [x for x in lst if x >= 0]
negative = [x for x in lst if x < 0]

print("Positive numbers:", positive)
print("Negative numbers:", negative)
```

2) Write a Python function to check whether a number is perfect or not.

```
def is_perfect(n):
    total = 0
    for i in range(1, n):
        if n % i == 0:
            total += i
    return total == n

num = int(input("Enter a number: "))
if is_perfect(num):
    print(num, "is a Perfect Number")
else:
    print(num, "is not a Perfect Number")
```

OR

2) Write a python program Check if a Substring is Present in a Given String.

```
string = input("Enter a string: ")
substring = input("Enter substring to check: ")

if substring in string:
    print("Substring is present")
else:
    print("Substring is not present")
```

Slip 25

1) Write a Python Program to Check Armstrong Number

```
num = int(input("Enter a number: "))
order = len(str(num))
sum = 0
temp = num

while temp > 0:
    digit = temp % 10
    sum += digit ** order
    temp //= 10

if sum == num:
    print(num, "is an Armstrong Number")
else:
    print(num, "is not an Armstrong Number")
```

2) Write a Python program to accept n elements in a list and Find the largest and smallest element also display the sum and average of all elements.

```
n = int(input("Enter number of elements: "))

lst = [int(input(f"Enter element {i+1}: ")) for i in range(n)]

largest = max(lst)
smallest = min(lst)
total = sum(lst)
average = total / n

print("Largest element:", largest)
print("Smallest element:", smallest)
print("Sum of elements:", total)
print("Average of elements:", average)
```

OR

2) Write a Python program to create a dictionary from a string.

Sample String: "Programming" . **Expected output:**{'p':1,'o':1,'r':2, 'a':1,'m':2, 'i':1,'g':2}

```
s = "Programming".lower()

d = {}
for ch in s:
    if ch in d:
        d[ch] += 1
```

```
else:
    d[ch] = 1

print(d)
```

Slip 26

1) Write a Python program to sum all the items in a list.

```
lst = [int(x) for x in input("Enter elements: ").split()]

total = sum(lst)

print("Sum of all elements:", total)
```

2) Write a Python function to check whether a number is in a given range.

```
def check_in_range(num, start, end):
    return start <= num <= end

number = int(input("Enter a number: "))
range_start = int(input("Enter range start: "))
range_end = int(input("Enter range end: "))

if check_in_range(number, range_start, range_end):
    print(number, "is in the range")
else:
    print(number, "is not in the range")
```

OR

2) Write a Python program to create a list of tuples with the first element as the number and second element as the square of the number.

```
n = int(input("Enter number of elements: "))

lst = [(i, i**2) for i in range(1, n+1)]

print(lst)
```

Slip 27

1) Write a Python Program to Check if a Number is Odd or Even

```
num = int(input("Enter a number: "))
```

```

if num % 2 == 0:
    print("Even Number")
else:
    print("Odd Number")

```

2) Write a Python function that accepts a string and calculate the number of upper case letters and lower case letters.

```

def count_upper_lower(s):
    upper_count = 0
    lower_count = 0
    for char in s:
        if char.isupper():
            upper_count += 1
        elif char.islower():
            lower_count += 1
    return upper_count, lower_count

input_string = input("Enter a string: ")
upper, lower = count_upper_lower(input_string)
print("Number of uppercase letters:", upper)
print("Number of lowercase letters:", lower)

```

OR

2) Write a Python program to accept a list of integers and partition it such that: All negative numbers appear first then zeros then positive numbers.

```

lst = [int(x) for x in input("Enter integers: ").split()]

negatives = [x for x in lst if x < 0]
zeros = [x for x in lst if x == 0]
positives = [x for x in lst if x > 0]

partitioned = negatives + zeros + positives

print("Partitioned list:", partitioned)

```

Slip 28

1) Write a python program to perform arithmetic operations (+, -, *, /, %).

```

a = float(input("Enter first number: "))
b = float(input("Enter second number: "))

addition = a + b

```

```

subtraction = a - b
multiplication = a * b
division = a / b if b != 0 else "Undefined (division by zero)"
modulus = a % b if b != 0 else "Undefined (modulus by zero)"

```

```

print("Addition:", addition)
print("Subtraction:", subtraction)
print("Multiplication:", multiplication)
print("Division:", division)
print("Modulus:", modulus)

```

2) Write a Python program to create a union and an intersection of two sets.

```

set1 = {int(x) for x in input("Enter elements of first set: ").split()}
set2 = {int(x) for x in input("Enter elements of second set: ").split()}

```

```

union_set = set1 | set2
intersection_set = set1 & set2

```

```

print("Union:", union_set)
print("Intersection:", intersection_set)

```

OR

**2) Write a Python script to concatenate following dictionaries to create a new one. Sample Dictionary: dic1={1:11,2:22} dic2={3:33,4:44} dic3={5:55,6:66}
Expected Result: {1: 11,2: 22, 3:33, 4: 44, 5: 55,6: 66}**

```

dic1 = {1: 11, 2: 22}
dic2 = {3: 33, 4: 44}
dic3 = {5: 55, 6: 66}

```

```

new_dic = {}

```

```

for key in dic1:
    new_dic[key] = dic1[key]

```

```

for key in dic2:
    new_dic[key] = dic2[key]

```

```

for key in dic3:
    new_dic[key] = dic3[key]

```

```

print("Concatenated Dictionary:", new_dic)

```


Slip 29

1) Write a Python Program to Check if a Number is Positive, Negative or Zero.

```
num = float(input("Enter a number: "))

if num > 0:
    print("Positive Number")
elif num < 0:
    print("Negative Number")
else:
    print("Zero")
```

2) Copy element 22 and 66 from the following tuple in to a new tuple and display new tuple. tuple1 = (11, 22, 33, 44, 55, 66)

```
tuple1 = (11, 22, 33, 44, 55, 66)

new_list = []

for i in range(len(tuple1)):
    if tuple1[i] == 22 or tuple1[i] == 66:
        new_list.append(tuple1[i])

tuple2 = tuple(new_list)

print("New tuple:", tuple2)
```

OR

2) Write a Python program to convert lowercase letters to uppercase and vice versa.

```
s = input("Enter a string: ")
converted = ""

for ch in s:
    if ch.islower():
        converted += ch.upper()
    elif ch.isupper():
        converted += ch.lower()
    else:
        converted += ch

print("Converted string:", converted)
```

Slip 30

1) Write a python program to find the frequency of characters in a string.

```
s = input("Enter a string: ")

freq = {}

for ch in s:
    if ch in freq:
        freq[ch] += 1
    else:
        freq[ch] = 1

print("Frequency of characters:", freq)
```

2) Write a program to count the number of items in a dictionary without using len().

```
d = {'a': 10, 'b': 20, 'c': 30, 'd': 40}

count = 0
for key in d:
    count += 1

print("Number of items in the dictionary:", count)
```

OR

2) Write a Python program to define a function that checks whether a number is prime or not. The function should return True or False.

```
def is_prime(n):
    if n <= 1:
        return False
    for i in range(2, int(n**0.5)+1):
        if n % i == 0:
            return False
    return True

num = int(input("Enter a number: "))

if is_prime(num):
    print(num, "is a Prime Number")
else:
    print(num, "is not a Prime Number")
```